- Sequencing bias & errors
- Speed & RAM
- Data storage & visualization
- Statistics

# Specific challenges:
# (3) transcriptomics

# Batch effect

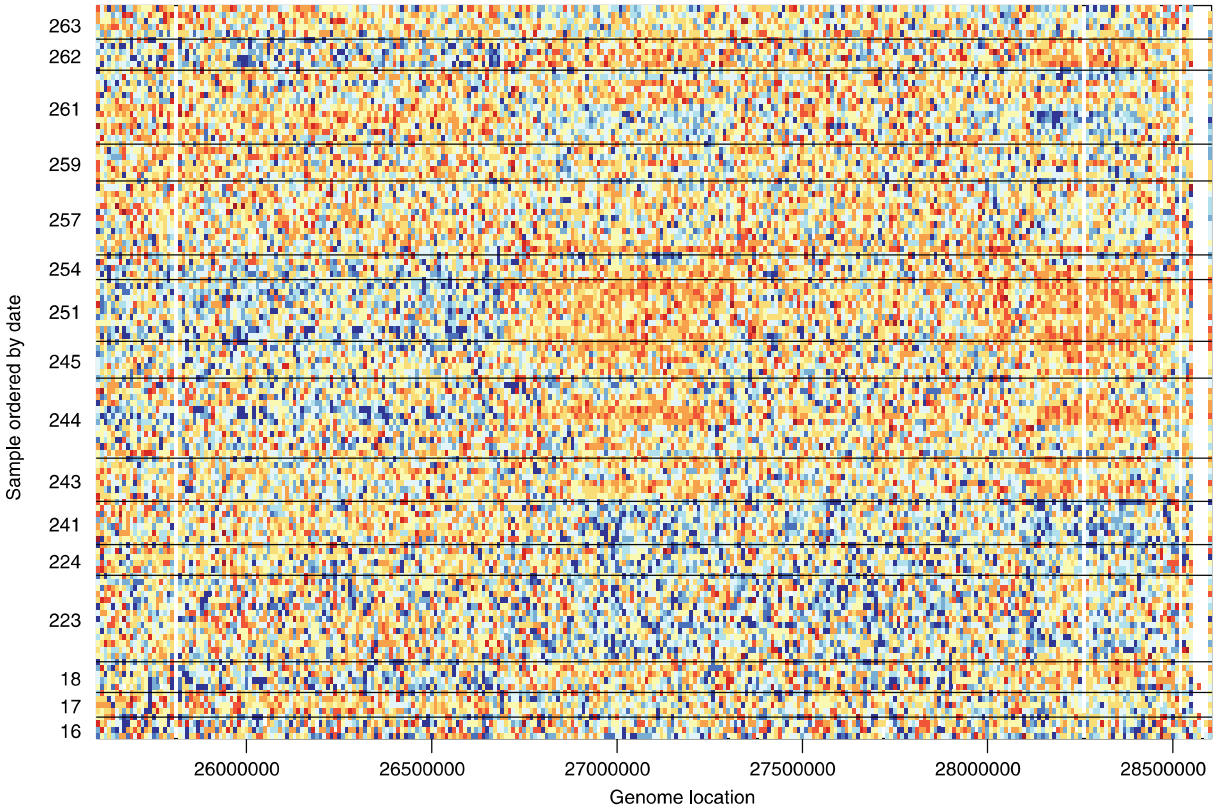
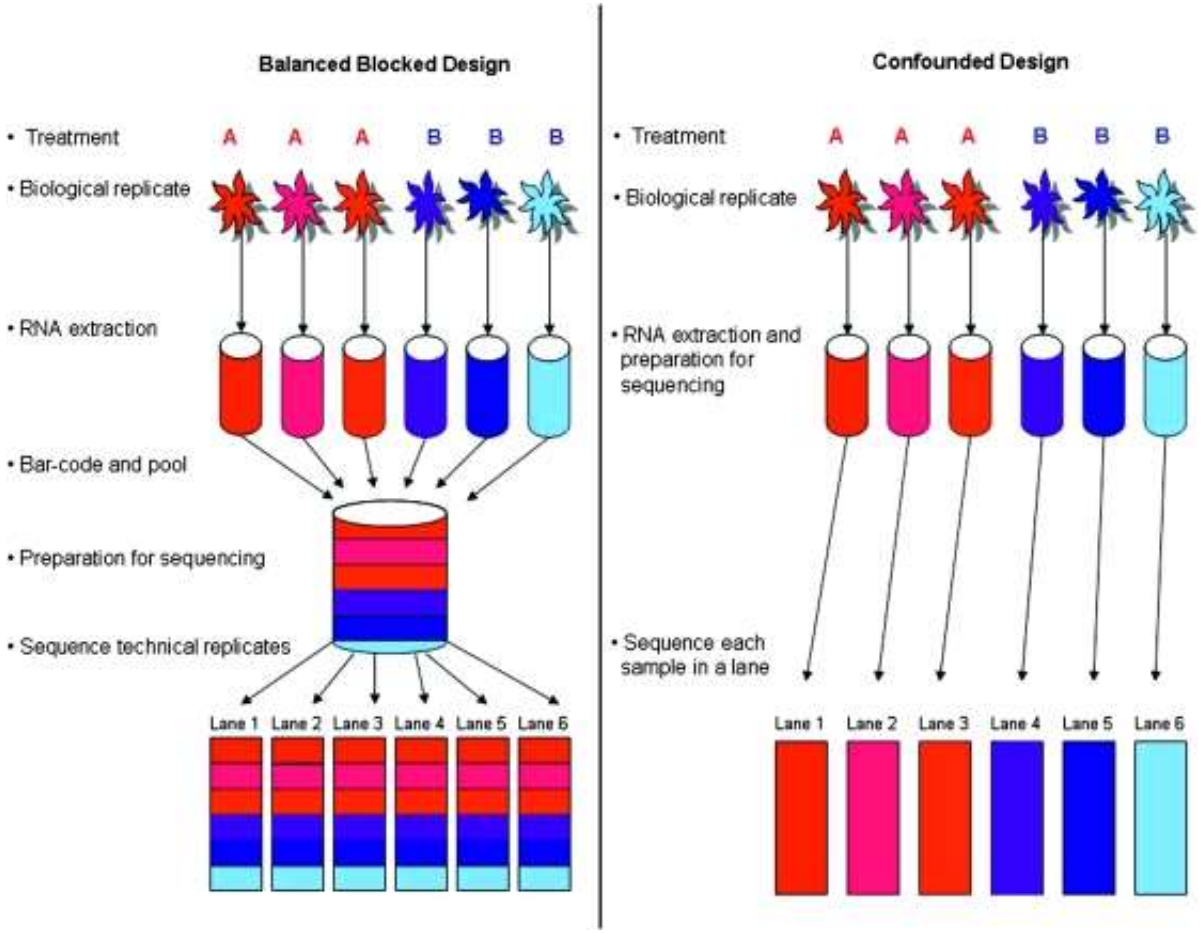



**Figure 3. Batch effect for second-generation sequencing data from the 1000 Genomes Project.** This figure is similar to one from Leek *et al.* [10]. Each row in the heat-map is data from a different HapMap sample processed in the same facility with the same platform (see Leek *et al.* [10] for a description of the data), shown for a 3-Mb region on chromosome 16, with data summarized in 10-kb bins. Data from each bin were standardized across samples, with blue representing 3 standard deviations below average, and orange representing 3 standard deviations above average. The rows are ordered by date, with black lines separating different processing days. The largest batch effect can be seen on the alternating pattern of blue and orange on days 223 to 241 and days 244 to 251.

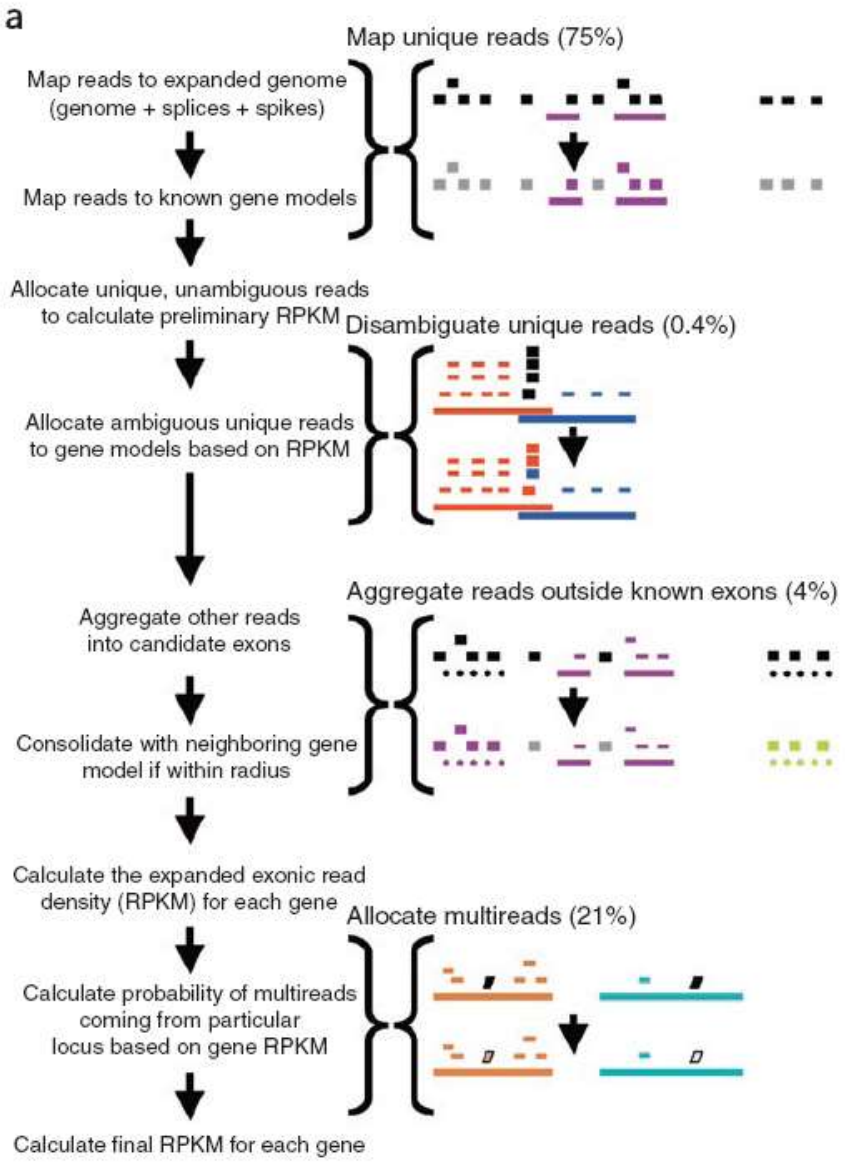
Taub et al. Genome Medicine 2010




Two designs for testing differential expression between treatments A and B.

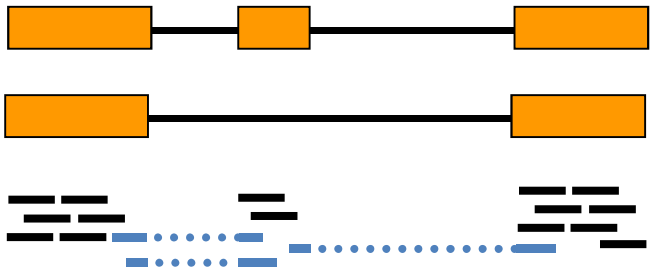
Auer and Doerge, Genetics 2010

# RNASeq – Transcript Complexity




Mapping :

- Reads with multiple locations
  - Conserved domains ?
  - Sequencing error ?
- Reads Spanning Exons
  - Gapped alignments ?
  - Sequencing error ?



Erange Pipeline : Mortazavi et al., Nature Methods 2008